MIL-C-19699D(NU)
10 April 1981
SUPERSEDING
MIL-C-19699C(SA)
19 March 1975

MILITARY SPECIFICATION

CLOTH, COATED (NYLON TAFFETA)

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for chloroprene coated nylon cloth.

1.2 Classification Coated cloth covered by this specification shall be of the following types:

Type I - Chloroprene coated
Type II - Chloroprene coated, flame resistant

2. APPLICABLE DOCUMENTS

2.1 Issue of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging and Packing of Coated (Plastic Rubber) and Laminated Fabrics

MILITARY

MIL-C-21852 - Cloth, Taffeta, Nylon

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing & Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580).

3. REQUIREMENTS

3.1 Guide sample Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 6.2 and 4.3).

3.3 Materials

*3.3.1 Base cloth The base cloth shall conform to type III of MIL-C-21852. The color for type I shall match Green Shade 3406. The color for type II shall be natural (undyed) and the colorfastness requirements shall not apply.

3.3.2 Coating compound The coating compound shall be chloroprene rubber that is free from objectionable odor, plasticized and pigmented black; and when used for the type II cloth shall be formulated with flame proofing additives. No elastomer other than chloroprene shall be used.

3.3.3 Coated cloth The coating compound shall be applied to one side of the base cloth.

3.4 Physical requirements The coated cloth shall conform to the requirements specified in Table I when tested as specified in 4.5.

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Table I - Physical Requirements

Test	Requirements Type I	Type II
Weight, (ounces per sq. yd.)	4.7 (± 0.5)	5.5 (± 0.5)
Breaking strength (lbs) (minimum)		
Warp	110	110
Filling	95	95
Tearing strength (grams) (minimum)		
Warp	850	800
Filling	650	600
Hydrostatic resistance (lbs/sq in) (min)		
Initial	80	80
After 5 launderings 1/	30	-
After 3 dry cleanings 1/	60	-
After 40 hours weatherometer 1/	60	-
Flex stiffness (inch-lbs x 10^{-4}) (max)		
At 70° F		
Warp	2.5	3.0
Filling	1.5	2.0
At -40°F		
Warp	8.0	-
Filling	4.5	-
At 0°F		
Warp	-	8.5
Filling	-	5.0
Adhesion of coating (lbs/2 inch width) (min)	10	10
Flame resistance, vertical (warp and filling)		
After flame time, seconds (coated side only) (maximum)	-	10.0
Char length, inches (maximum)	-	8.0
Resistance to heat at 160°F	No evidence of tackiness, exudatio or loss of flexibility	
Blocking (scale rating) (max)	No. 2	No.2

1/ After subjection to laundering, dry cleaning or weathering, but prior to hydrostatic testing, the specimens shall show no evidence of cracking, crazing, blistering, peeling, swelling or tackiness.

3.5 Dusting medium The coated cloth shall be dusted with a suitable dusting medium that does not support mildew growth. Only the minimum amount of dusting powder necessary to prevent blocking should be applied. The application of dusting powder shall be controlled during processing or shall be reduced by an after processing brush or wash so that the coated side of the cloth shows not less than "good" resistance to crocking when tested as specified in 4.5.

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3.6 Length and put up The coated cloth shall be put up on rolls as specified in PPP-P-1136. Alternately, the coated cloth may be put up on rolls in 120 to 160 yards lengths with no more than 4 pieces per roll and no piece less than 25 yards in length.

3.7 Width The finished width of the coated fabric, including the selvage, shall be as specified (see 6.2).

3.8 Workmanship The finished coated cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

*4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection When required, the first article submitted in accordance with 3.2, shall be visually inspected for color, finish and appearance and shall be tested for the chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.4.1 Component and material inspection In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specification, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition, the contractor shall furnish a certificate of compliance for the coating compound formulation requirement specified in 3.3.2 and for the non-mildew supporting characteristic of the dusting medium specified in 3.5.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 and 4.4.2.2.

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4.4.2.1 Yard-by-yard examination The required yardage of each roll shall be inspected on one side only. However, the side shall be alternated, so that alternate rolls are examined on the face and back respectively. The defects found shall be classified as listed below. In addition, the cloth shall be given a through-light inspection for pinholes. The through-lighting inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted, regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. The lot size shall be expressed in units of one linear yard. The sample unit for this examination shall be one linear yard. The acceptable quality levels (AQL's) for this examination shall be 1.5 for major defects and 4.0 for total defects (major and minor) per 100 units (yards). The inspection level shall be level II. The number of rolls from which the sample is to be selected shall be in accordance with Table II. An approximately equal number of yards shall be examined in each roll sampled.

Defect	Classification	
	Major	Minor
UNCOATED SIDE		
Any hole, cut, tear or smash	X	
Any slub or multiple float		X
Baggy, ridgy or wavy cloth	X	
Any strike through of coating		X
Color shaded or streaked		X
Any spot or stain		X
COATED SIDE		
Any cut, scratch, adraded area, hole or tear	X	
Any uncoated area	X	
Any area where coating is noticeably thinner	X	
Any blister, tunnel or delamination of coating	X	
Any lump or heavier coated area		X
Foreign matter imbedded in coating	X	
Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure	X	
Coating cracked or checked	X	
Coating not applied over full width of base cloth	X	
Any pinhole	X	
Any light area or window resulting from improper distribution of gipment		X
Coating streak		X
Any ripple, waviness or dimensional distortion	X	
Any spot, stain or streak		X
Width not as specified		X
Color not as specified, shaded or streaked		X
Any evidence of uncleanliness		X
Any tackiness	X	
Any objectionable odor (other than that which is characteristic of the coating compound)		X

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Table II - Sample size

Lot Size (yards)	Sample size
Up to 1200 inclusive 1/	3
1201 to 3200 inclusive	5
3201 to 10,000 inclusive	8
10,001 to 35,000 inclusive	13
35,001 to 150,000 inclusive	20
150,001 and over	32

1/ If the lot contains fewer than 3 rolls, each piece in the lot shall be examined.

4.4.2.2 Examination for length

4.4.2.2.1 Individual pieces During the yard-by-yard examination of the cloth, each piece in the sample shall be examined for length. Any length found to be less than the minimum or more than the maximum specified, or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.4.3 Examination of packaging An examination shall be made in accordance with the provisions of PPP-P-1136 to determine that packaging, packing and marking comply with Section 5 requirements of this specification.

4.5 Testing of the end item The methods of testing specified in FED-STD-191 wherever applicable, and as listed in Table III shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for test purposes shall be 4 continuous yards full width of cloth in the vulcanized state. The lot size shall be expressed in units of one yard. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

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Table III - Instruction for testing

Characteristic	Requirement reference	Test method
Weight	Table I	5041
Breaking strengthl		
Warp and Filling	Table I	5100
Tearing strength		
Warp and Filling	Table I	5132 1/
Hydrostatic		
Initial	Table I	5512 2/
After 5 launderings (cotton mobile)	Table I	5556 3/ and 5512 2/
After 3 dry cleanings	Table I	5508 3/ and 5512 2/
Weatherometer (after 40 hours)	Table I	5804 3/ 4/ and 5512 2/
Flex stiffness		
at 70°F		
Warp and Filling	Table I	5206 5/
At -40°F		
Warp and Filling	Table I	5206 5/ 6/
At 0°F		
Warp and Filling	Table I	5206 5/ 6/
Adhesion (pounds per 2 inch width)	Table I	5907
Resistance to heat	Table I	5850 7/
Flame resistance	Table I	5903
Blocking	Table I	5872
Resistance to crocking	3.5	5651 8/

1/ Standard equipment with augmenting weight.
2/ Water pressure shall be applied against fabric side.
3/ After exposure, the specimens shall be examined for evidence of cracking, crazing, blistering, peeling, swelling and tackiness.
4/ The coated side of the cloth shall be exposed to the carbon arc.
5/ The geometric mean of the coating up average and the coating down average shall be indicated. Five determinations with coating up and 5 determinations with coating down shall be made on each sample unit.
6/ The coated cloth shall be cooled for 1 hour at the specified temperature (± 2°F) prior to test.
7/ The cloth shall be exposed for 120 hours at 160°F and shall be examined for signs of tackiness, exudation or loss of flexibility when flexed or crumpled manually.
8/ The test shall be conducted using the blue crock cloth and shall be tested for resistance to dry crocking only.

5. PACKAGING

5.1 Preservation-packaging Packaging shall be level A or C (see 6.2).

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5.1.1 Levels A or C The coated fabric shall be packaged in accordance with the requirements of PPP-P-1136.

5.2 Packing Packing shall be level A, B or C (see 6.2).

5.2.1 Levels A, B or C The coated fabric shall be packed in accordance with the requirements of PPP-P-1136, except that 4 rolls may be packed for any type of container specified.

5.3 Marking In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with PPP-P-1136.

6. NOTES

6.1 Intended use The coated fabrics covered by this specification are intended for use in the manufacture of special purpose cold weather clothing worn by Navy personnel.

6.2 Ordering data Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Whether first article is required (see 3.2).
(d) Width required (see 3.7).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Guide sample For access to guide sample, address the procuring activity issuing the invitation for bids (see 3.1).

*6.4 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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